A Study on Multi-Stage Greywater Filtration Integrating Coconut Coir, Activated Carbon, and RO-UV Technologies: Treatment Efficiency and Non-Potable Reuse Feasibility

Sarvesh Singh[1], Dhananjay Yadav[2], Vijay Laxmi Singh [3], Krishna Pratap Singh [4], Pratish Kannaujiya[5*]

[1,2,3,4]UG Student, Department of Civil Engineering, Buddha Institute of Technology Gida Gorakhpur Uttar Pradesh, INDIA 273209

[5*]Assistant Professor, Department of Civil Engineering, Buddha Institute of Technology Gida Gorakhpur Uttar Pradesh, INDIA 273209

[1]sarveshsinghrajput895@gmail.com, [2]dhananjayyadav0958@gmail.com, [3]vijaydeep3127@gmail.com,

[4]krishnapratapsinghss123@gmail.com, [5*]kannaujiyapratish@gmail.com

Abstract: Greywater is 60-75 percent of household wastewater and is one of the easily accessible resources that can help save up to a large part of freshwater consumption in case they are treated with the help of effective, cost-efficient and sustainable systems. This paper offers the design, manufacture and overall performance analysis of a multi-stage greywater filtration unit that is developed as an integrated unit and has been designed to offer high purification efficiency with low-cost materials that can be found locally. The treatment train uses a primary sedimentation phase and sequential layers of filtration using cotton cloth to screen the dissolved organic compounds, activated charcoal to adsorb the dissolved organic molecules, coconut coir to entrap the dissolved organic compounds, and fine sand to remove turbidity and suspended solids after which the graduated gravel is used to maintain structure and control the flow of water. A series of polishing operations such as pre- and post-carbon cartridge, reverse osmosis (RO) membrane, ultraviolet (UV) disinfection, and the conditioning of the mineral was added in order to improve the final effluent quality. Extensive physicochemical profile of the greywater samples taken at the source of bathroom and basin showed significant improvement in all the important parameters. Turbidity dropped by 58NTU to 6.5NTU (88.79per cent removal), total dissolved solids (TSS) dropped by 950mg/L into 480mg/L (49.47per cent removal), biochemical oxygen demand (BOD) removed 120 mg/L into 25mg/L (79.16per cent removal), chemical oxygen demand (COD) removed 280mg/L into 65mg/L (76.78per cent The dissolved oxygen rose to 6.4 mg/L and the hardness dropped to 190mg/L (53.65% removed) and the pH reached 7.05 which is in the allowable range of 6.5 to 8.5 to make it non-potable reuse. These results are close to those performance ranges that have been reported in hybrid filtration systems, in granular media filters, in biochar-based adsorptive units, and in decentralized nature-based greywater treatment systems. The developed system resulted in transparent, odourless and highly purified greywater, which has met the recommended limits in reuse non-drinking purpose, including irrigation, toilet flushing, landscape and cleaning. This research paper proves that the combination of physical, biological, and adsorption-based processes can achieve high quality treated greywater with significantly low operational cost in the form of a hybrid filtration geometry. This system is an efficient, sustainable and practical solution to decentralized greywater management, especially in rural and resource-heavy areas whereby access to the traditional treatment infrastructure is constrained due to its simplicity, affordability and scalability.

Keywords: *Greywater treatment, Multi-stage filtration, Domestic wastewater recycling, Sand and gravel filtration, Activated carbon adsorption, Biochar based filtration, Hybrid treatment system, Physicochemical analysis, Enhancement of water quality, Sustainable water reuse, Decentralized wastewater management, Low-cost treatment technologies, Household wastewater purification, Environmental sustainability.*

INTRODUCTION

Water shortage is a serious international problem that has been aggravated by high rates of urbanization, global warming and increased domestic water consumption. Greywater, which is waste water produced when hand washing, bathing, laundry and kitchen are carried out, constitutes 60-75% of domestic wastewater in regular households (Oteng-

Peprah et al., 2018), hence providing a great potential in decentralized water reuse. Greywater should be used and reused through effective treatment methods that can significantly help to save the freshwater use especially in non-portable uses such as irrigation, toilet flushing, laundry and general landscape management (Oteng-Peprah et al., 2018). The nature of greywater differs greatly based on household operations and socioeconomic status, yet typically includes suspended solids, detergents, oils and fats, nutrients, pathogens, and complex organic matter that need to be systematically treated to fit the set reuse requirements (Oteng-Peprah et al., 2018). Among various technologies of treating greywater, granular media filtration has received a lot of research attention because of its relatively simplicity, low capital and operational expenses, and proven high rate of pollutants removal. A comprehensive literature review of the research by Shaikh and Ahammed (2022) highlighted that media properties such as sand grain size, gravel gradation, and charcoal porosity have a considerable impact on removal performance among the key parameters such as turbidity, BOD, COD, and microbiological contaminants. Zipf and Hinz (2016) used laboratory and pilot-scale tests that showed that slow sand filtration with granular polishing media was effective in the elimination of suspended solids, organic load, and aesthetic quality, proving the effectiveness of this type of systems in household-scale greywater treatment. Equally, Spychala et al. (2019) indicated that the layered sand filters yielded significant solids and organic matter removals, which also support the use of the treatment method in domestic greywater treatment settings. Simultaneously, there exist the developments of hybrid and optimized filtration systems that have made the treatment results more reliable and effective. Akbari et al. (2025) constructed a multi-stage system with the integration of coagulation, sand, and activated carbon filtration with optimization showing that the combination of the media and systematic depth of the layer has enhanced the removal of the pollutants. Hybrid systems comprising of electrocoagulation and filtration have also improved the prevention of more complicated contaminants and persistent organic pollutants. According to Waghe et al. (2024), a set-up consisting of electrocoagulation (EC) with further filtration using sand and activated carbon adsorption recorded tremendous COD, colour and turbidity reductions. Devikar et al. (2021) reported a complementary solar-assisted hybrid system which indicates that EC-filter units powered by renewable energies are especially acceptable for rural and off-grid locations where the power supply availability was restricted. The use of biochar-based filtration technologies has increased recently due to the high adsorption capacity of biochar, high surface area, and low environmental footprint. The Quispe et al. (2023) biochar filters were optimized to treat handwashing greywater with an objective of revealing significant enhancement of the biochar filters using systematic manipulations of media size and contact time to achieve better removal of BOD, COD, and turbidity. Further detailed reviews by Muoghalu et al. (2023) affirm the multifunctional process of biochar removal such as physical adsorption, ion exchange, and catalyzing microbial degradation, making biochar a good candidate as a sustainable decentralized filtration system. Constructed wetlands along with other nature-based treatment solutions are low-energy and sustainable greywater recovery options. Avery et al. (2007) and Arden et al. (2018) showed how horizontal and vertical-flow built wetlands are capable of eliminating nutrients, pathogens, and organic pollutants by the action of sedimentation, plant uptake, and biodegradation by microbes. The broader applications like constructed wetland microbial fuel cells (CW-MFCs) have been shown to have a potential of increasing the pollutant degradation and at the same time produce small amounts of electricity. The investigations by Hartl et al. (2021, 2020) also confirmed the enhanced elimination of micropollutants and increased bacteria activity through wetland-based bioelectrochemical system. Also, the behaviour of long-term hydraulic regimes and the holistic design of wetlands, based on systems, is covered by Hartl et al. (2022). On a more systemic level, greywater reuses are seen as a more and more effective solution that can be used to create urban water resilience. As Van de Walle et al. (2023) stressed, the use of decentralized greywater systems will lessen the pressure on the centralized municipal infrastructure to the same extent as the overall water circularity will increase. Such community-scale case studies as conducted by Masoud et al. (2025) do show that nature-based greywater treatment systems can be socially acceptable, environmentally beneficial and economically sustainable, especially in arid locations that have limited water. According to the principles of science, which are presented in these works, the current study is devoted to the constructive design and testing of a multi-stage greywater filtration station with the usage of cheaper natural resources (gravel, sand, coconut husk, activated charcoal) and the use of high-technology post-treatment units (RO + UV + mineral cartridge). System design complies with the hybrid and multi-layer systems which are described by Akbari et al. (2025), filtration dynamics reported by Shaikh and Ahammed (2022), adsorption mechanisms reported by Quispe et al. (2023), and natural filtration principles reported by Arden et al. (2018).

The main objective of the research papers are to develop an economical, efficient, and sustainable greywater treatment system to be applied in a household or a small community setting. Assess the performance of the system on the basis of major physicochemical and microbiological parameters. Show the treated effluent is fit to meet the recommended standards of non-portable reuse applications. Confirm that resource efficient, economical, and sustainable filtration designs can indeed improve the quality of greywater and can be used to sustain water conservation procedures especially in environments with limited resources. The system design is focused on simplicity, low energy use, simplified maintenance, and use of locally available materials- aspects that the global greywater research community highly endorses and that are well in line with United Nations Sustainable Development Goal 6 (Clean Water and Sanitation).

2. MATERIALS AND COMPONENETS

2.1 Filter Media

The multi-stage greywater treatment system made use of a mixture of both natural and engineered filter media that was chosen on the basis of the ability to remove all types of contaminants that are present, including the physical, chemical and biological ones. The main medium of filtration was fine sand with a particle size of 0.2-0.6 mm because it has a high specific surface area and uniform pore structure, and this makes it an effective filtration medium against suspended solids, colloidal particles, and turbidity. The sand layer also favored the growth of biomasses which aided to a certain degree in the biological decay of organic matter. A layer of sand was covered with graded gravel (4-8 mm and 12-20 mm) and pebbles (20-30 mm) to enhance hydraulic conductivity, even distribution of flow, inhibit media migration and support the mechanical integrity of the layer besides providing aeration and fixation of microbes. Activated carbon (iodine number of 900-1100 mg/g) with a specific surface area of 900-1200 m2/g was added to adsorb dissolved organic matter, surfactants, colour, odour, residual chlorine, and trace contaminants by using its micro- and mesoporous structure (particles size 0.5- 2 mm). The coir (coconut husk) is a lignocellulosic tissue with fibre length of 10-25 cm and lignin content of 40-45 and was used as an additional natural medium to increase the adsorption of oils and fine particulates, pH buffering, microbial activity, and sustainability of the overall system. The details of all media filter treatment as shown in Figure 1.

Fig 1. Illustrates the details of the multimedia filter treatment system for primary treatment.

2.2 Treatment Units and Auxiliary Equipment:

A 24 V DC booster pump (operating pressure: 60-100 psi; flow rate: 1-2 L min-1) was attached to keep the hydraulic conditions constant throughout the treatment train and stabilize the work of pressure-depending units, in particular, the reverse osmosis (RO) membrane. Solids pre-treatment was performed with the help of a sedimentation chamber made of HDPE/PVC with a hydraulic hold time of 45-60 min, so that the gravitational settling of grit and heavy particulates would occur, and the solids load on the further filtration steps would be lessened. To reduce fouling and increase the RO membrane service life, a 5 um pre-filter that used activated carbon was utilized to take out coarse organic material, volatile organic compounds and chlorine left behind. The polishing after treatment was also done using 1-5 um activated carbon filter to clear off odour, chlorine and other trace organic pollutants. A mineral cartridge filter (calcium carbonate ($CaCO_3$), magnesium oxide (MgO) and mineral stones were used to restore minerals and improve pH stability, electrical conductivity and mineral content of the treated effluent by activating the device at a flow rate of 0.5-1 L min-1. Further purification was done with an RO membrane whose nominal pore size was 0.0001 um with a total dissolved solids rejection efficiency of 90-98 and a range of operating pressure of 60-100 psi which allowed efficient removal of dissolved salts, heavy metals, nutrients, detergents and pathogenic microorganisms. Last disinfection was done with a UV unit at 254 nm (UV-C) of power of 8-11 W and a flow rate of 1 L min-1 without adding a chemical to inactivate bacteria, viruses, and protozoa. Interconnections between the treatment units were done using food-grade PVC flexible hose pipes (1/2-inch diameter) to make sure that they will not leak. A pre-filter high-porosity polyurethane sponge (2-5 cm thick) was installed at the system inlet to clean up coarse impurities and floating debris, which causes clogging in the downstream units. The digital pH meter (range: 0-14; accuracy: +-0.01) was used to monitor water quality to ascertain whether the treated greywater had acceptable standards of reuse (pH 6.5-8.5). The details of all media filter as a primary treatments as shown in Figure 2.

Fig 2. Schematic representation of the treatment units and auxiliary equipment used in the proposed greywater treatment system.

3. METHODOLOGY

3.1 System Design and Greywater Collection

Greywater was pooled in domestic shower facilities and washbasin outflows and these were typical household wastewater that included surfactants, oils, suspended solids, colloidal matter and biodegradable organic compounds. The samples were brought using clean and sterile and airtight containers and taken to be treated immediately to ensure minimization of physicochemical changes and microbial deterioration before the process. The treatment system was made into a vertically oriented multi stage filtration column that was fabricated with chemically inert polyvinyl chloride (PVC) and high-density polyethylene (HDPE). The column was made of successive layers of filter material (also from top to bottom) namely, sponge pre-filter (2-5 cm thickness), cotton cloth, layer of activated charcoal, coconut coir, fine sand (15-20 cm depth), medium gravel (4-8 mm; 10-15 cm), coarse gravel (12-20 mm; about 10 cm), and layer of pebbles (20-30 mm; 5-10 cm) as shown in Figure 3. The chosen configuration was derived in order to allow the removal of contaminants in a progressive manner using the physical straining method, adsorption and the biological method. The filter media were washed in time with distilled water to get any surface impurity off and dried at 105℃ over 24 h and sifted using standard mesh sizes in order to get the same distribution of the particles and uniform pores. Flow through the system was controlled by either gravity or by a booster pump to ensure that the hydraulic loading conditions in the system were constant and that there was no disturbance or compaction of the filter layers.

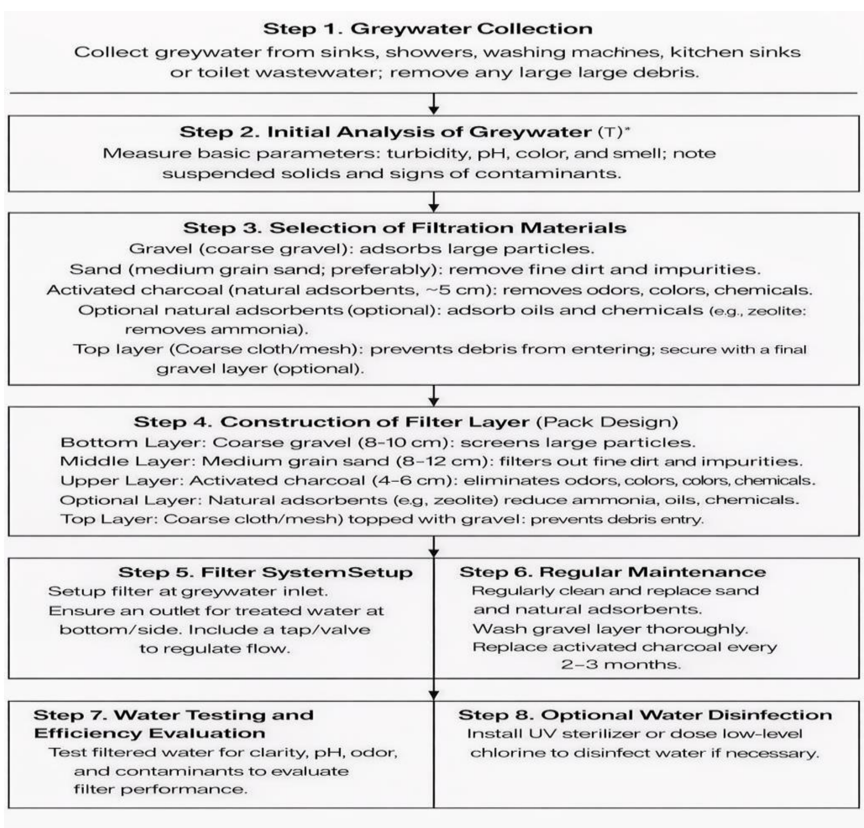

Fig 3. Schematic flowchart of the greywater treatment methodology using a multilayer filtration system.

3.2 Multi-Stage Treatment Configuration

In this step, the treatment is broken into many stages, each with a particular goal. Raw greywater was first used as a pre-treatment step in which the raw greywater was exposed to a dedicated sedimentation tank with a hydraulic retention time of 45-60 min. It allowed the elimination of heavy particles, grit and settable solids, and thus, it decreased turbidity and decreased the load of the particulate matter to the downstream treatment units. After the sedimentation the supernatant was subjected to multi-layer filtration column where the contaminants were eliminated by a combination of depth filtration, adsorption, and biodegradation using biofilm. Physical filtration took place mainly in the layers of sand and gravel, and the chemical adsorption was supported by the use of activated carbon and coconut coir, which boosted biological activity. The effluent of the first filtration column was then processed in a series of polishing and advanced purification unit, which includes pre-carbon cartridge adsorbing the residual organic compounds and volatile organic compounds, reverse osmosis (RO) membrane removing dissolved salts, ionic species and inorganic contaminants and ultraviolet (UV) disinfection unit eliminating microorganisms. A post-carbon cartridge was used to give final polishing to enhance aesthetic quality and a mineral cartridge was used to replenish important minerals and stabilize the pH. This combined multi-barrier mode of treatment guaranteed full elimination of physical, chemical, and microbiological contaminants that are found in greywater.

3.3 Sampling and Analytical Methods

Water samples were taken at four important points of the treatment process; (i) raw greywater influent, (ii) post sedimentation effluent, (iii) effluent of the filtration column, and (iv) final treated water after polishing units. All the samples were examined as per the standard procedures as required by the American Public Health Association (APHA) and applicable Indian Standards (IS). The main water quality parameters, pH, turbidity, total dissolved solids (TDS), biochemical oxygen demand (BOD_5), chemical oxygen demand (COD), dissolved oxygen (DO), and total hardness were identified at the standard laboratory parameters. The efficiency of removal of each parameter (e) was obtained as follows:

$$\eta(\%)= \frac{C_{in}-C_{out}}{C_{in}} \times 100$$

The influent and effluent concentration are represented as C_{in} and C_{out} respectively. All the analyses have been done thrice and reported in the form of mean values and standard deviations are attached.

4. EXPERIMENTAL RESULT

The experimental study was done to test the efficiency of developed multi-layer greywater treatment system which includes sand, gravel, activated charcoal, coconut husk, carbon filtration units, reverse osmosis (RO) and ultraviolet (UV) disinfection. Greywater samples at various treatment stages were taken and the essential physicochemical parameters evaluated to determine the efficiency of treatment. The outcomes that were achieved prior to the filtration were compared to the acceptable levels of reuse of non-drinking water.

4.1 pH Analysis

A very important parameter that impacts the water quality, biological activity and reuse suitability is pH. The PH of raw greywater was recorded at 6.83 which shows that the conditions were slightly acidic owing mainly to the presence of soaps, detergents and organic matters. Following intervention by the multi-stage filtration system, the pH had gone up to 7. 05 which is within the acceptable range of 6. 5-8. 5 in non-portable reuse applications. Such an improvement means that the filter media, especially activated charcoal, gravel, and coconut husk, are able to buffer. A moderate reduction in pH was observed in the fine sand layer, as a result of alkaline component adsorption, but in charcoal and gravel layers, buffering occurring, the near-neutral conditions were reestablished. The pH measurements at the various individual stages of filtration are shown in Table 4.1 which indicates how each medium contributes towards stabilization of pH.

Table 4.1 presents the pH values observed at different individual filtration stages, highlighting the contribution of each medium to pH stabilization.

Sl. No	Sample	pH values
1	Grey Water sample	6.84
2	Fine Sand sample	5.85
3	Charcoal sample	6.96
4	Gravel sample	6.93
5	Pre sample	6.62

4.2 pH Analysis for Combined Filter Media Layers

In order to determine the effects of synergistic media combinations to be used in the study, pH tests of water samples that were sent through various media combinations were performed. Table 4.2, which summarized the results, shows that the combination of activated charcoal gave a near-neutral pH value. The findings show that the combined effects of charcoal and coconut husk improve the buffering capacity, which guarantees that the pH level is maintained steadily in the treated effluent. Table 4.2 PH of various combinations of filter media.

Table 4.2 pH values for different combined filter media

Sl. No	Sample	pH values
1	Fine Sand & Gravel	6.84
2	Fine Sand & Charcoal	6.94
3	Charcoal & Gravel	6.97
4	Gravel & Coconut husk wood mixture	6.91
5	Charcoal & Coconut husk wood mixture	6.93

4.3 Water Quality Parameters Before and After Filtration

Table 4.3 compares the main water quality parameters prior to and after filtration. There were considerable changes in turbidity, total dissolved solids (TDS), biochemical oxygen demand (BOD), chemical oxygen demand (COD), as well as hardness, and a great deal of dissolved oxygen (DO). As per relevant standards, permissible limits of non-portable reuse. The decline in turbidity indicates that depth filtration is effective in removing suspended solids whereas fall in both BOD and COD is a confirmation that biodegradable and non-biodegradable organic matter is also removed. The rise in dissolved oxygen indicates a better water quality and a lesser organic load.

Table 4.3 Comparison of water quality parameters before and after treatment

Sr. No.	Parameter	Unit	Before Filtration (Sample A)	After Filtration (Sample B)	Permissible Limit (for Non-potable Use)*
1	pH	–	6.83	7.04	6.5 – 8.5
2	Turbidity	NTU	58	6.47	<10
3	TDS	mg/L	950	417	<2000
4	BOD	mg/L	120	24	<30

5	COD	mg/L	280	61	<100
6	DO	mg/L	2.1	6.1	>4
7	Hardness	mg/L	410	186	<500

4.4 Filtration Efficiency

General filtration efficiency of the constructed multi-stage greywater treatment system was measured in terms of the retention of the major physicochemical and organic pollution indicators. It is seen that the system attained a 88.79 percent turbidity reduction, which is evidence of the system successfully removing suspended and colloidal solids via depth filtration in the sand and gravel layers. The total dissolved solids (TDS) reduced by 49.47% which was due to adsorption by activated carbon and coconut husk and separation by the membrane in the reverse osmosis unit. Combined biological degradation, physical filtration, and adsorption of biodegradable and non-biodegradable organic material resulted in the reduction of biochemical oxygen demand (BOD5) and chemical oxygen demand (COD) by 79.16% and 76.78 respectively. The total hardness was reduced by 53.65 percent indicating that the divalent ions were successfully removed by adsorption and membrane treatment. Moreover, the level of dissolved oxygen (DO) rose substantially as the concentration of the same changed to 6.4 mg/L of treated water in comparison with 2.1mg/L of the untreated greywater which shows that the water quality was improved, and the organic load was decreased. These findings indicate that multi-layered treatment system comprising of physical, chemical, biological, membrane-based and disinfection processes are effective in improving the quality of greywater, making the treated effluent viable to be used in non-potable reuse processes, which include irrigation, toilet flushing and household cleaning.

5. CONCLUSION

This paper has been able to design, manufacture and test a multi-stage greywater treatment system, which involves inexpensive natural resources and technology of advanced polishing. The integrated system showed a significant and consistent enhancement of all the physicochemical parameters measured, with important reduction in turbidity (88.79%), TDS (49.47%), BOD (79.16%), COD (76.78%) and hardness (53.65%), without affecting the pH levels out of limits and with a great enhancement of the dissolved oxygen concentration. The outcomes of the experiment are consistent with the performance ranges found in the recent studies of multi-layer sand filtration, activated carbon adsorption, biochar-based treatment systems, electrocoagulation-assisted filtration, and decentralized nature-based greywater treatment technologies. The multi-stage filter design successfully reproduced the principles mentioned in the scientific literature: the granular filtration mode improves the removal of suspended solids, adsorption media enhances organic pollutants removal, and hydraulic residence time is a key factor that is relevant to the efficiency of the treatment. The hybrid designs are being mentioned in the literature of integrated coagulation-filtration and coagulation-adsorption designs that justify the improved results of this project, especially in terms of organic load and surfactant-related contaminants reductions. The resulting treated effluent was free of visual or odour effects, met quality standards recommended by the governments to reuse water that is not drinkable such as irrigation, toilet flushing and household cleaning. Addition of RO, UV and mineral polishing units also reinforced microbial and chemical quality which was critical in ensuring the safety and acceptability in the various reuse applications. In terms of sustainability and implementation, the greywater filtration system is one of the viable, low-cost and eco-friendly decentralized water reuse systems especially in areas with acute water scarcity and access to centralized treatment facilities. Its user-friendly nature characterized by the use of materials locally available and easy operation of the system implies that the system can be implemented in homes, small organizations and communities with small resources. This system with proper design maintenance procedures and regular checks can be successfully used in home and institutional environments to encourage the use of water responsibly and in the principles of a circular economy. The project enjoys national and global precedents, being a functional model of sustainable resource management such that the national and global projects like the Jal Jeevan Mission in India and UN SDG 6 national goals to conserve fresh water and reduce discharge of wastewater and ensure that the future generation will be able to sustain their environment.

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